Exp No: 5	Experiments on Packet capture tool: Wireshark

AIM:

To understand the features of Wireshark as a packet capture tool, to capture the packets passing through a network interface and to analyse the headers of the captured packets.

ALGORITHM / PROCEDURE:

1. Install and open Wireshark. The welcome screen lists all the network interfaces (Ethernet, Local Area Connection, VMware adapters, loopback adapter, USBPcap) with a small activity graph.
2. Select the active interface (**Ethernet**). A capture filter can be typed in the *using this filter* box before the capture starts, for example `arp`.
3. Double click the interface to start the capture. The live packets appear in the Packet List pane.
4. Study the columns of the Packet List pane: No., Time, Source, Destination, Protocol, Length and Info. The colour of a row shows the protocol (ARP, UDP, MDNS, SSDP, ICMPv6 ...).
5. Select one packet and expand its layers in the **Packet Details** pane (Frame, Ethernet II, Address Resolution Protocol).
6. Select a header field and see its bytes highlighted in the **Packet Bytes** pane (hexadecimal dump).
7. Stop the capture using the red square button. Display filters such as icmp, dns, tcp.port == 80 can be applied on the captured packets and the capture can be saved as a .pcapng file.

Filters used / protocols observed:

Item	Meaning
arp (capture filter)	Captures only the ARP packets
icmp, dns, tcp.port == 80 (display filters)	Show only ICMP, DNS and HTTP traffic from the captured packets
SSDP	M-SEARCH * HTTP/1.1 sent to 239.255.255.250 (device discovery)
MDNS	Standard query response sent to the multicast address 224.0.0.251
ARP	Who has 172.16.76.3? Tell 172.16.76.144 (broadcast request)
ICMPv6	Neighbor Solicitation sent by IPv6 hosts

DIAGRAM:

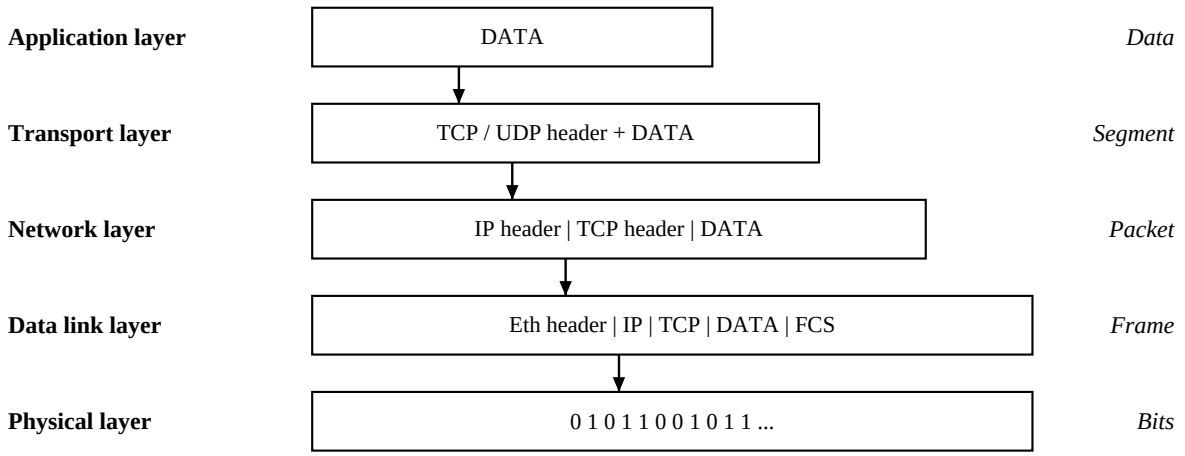


Fig 1: Encapsulation of data as it moves down the protocol stack

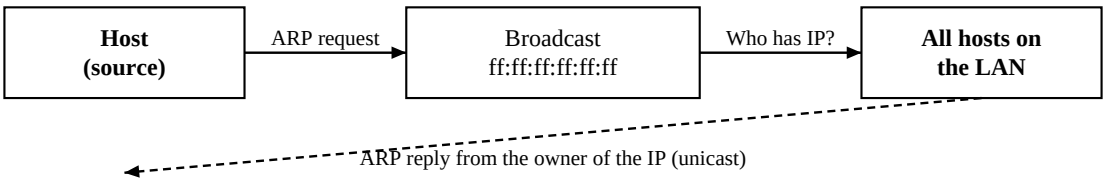
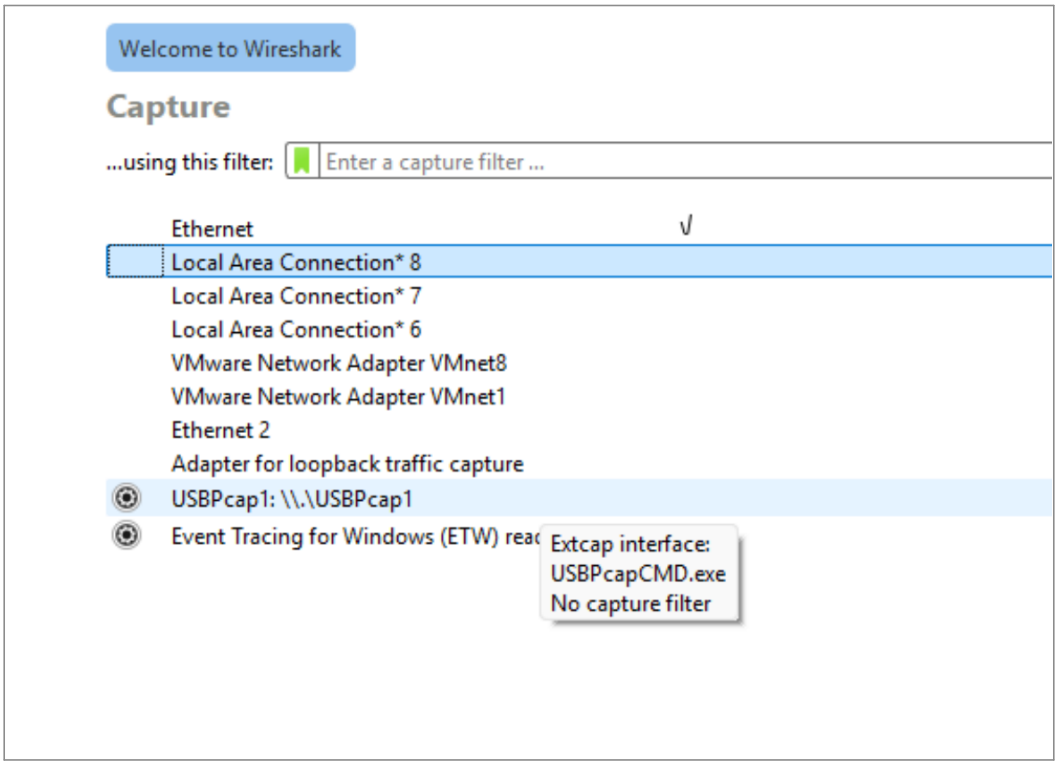


Fig 2: ARP request broadcast seen in the capture

OUTPUT:



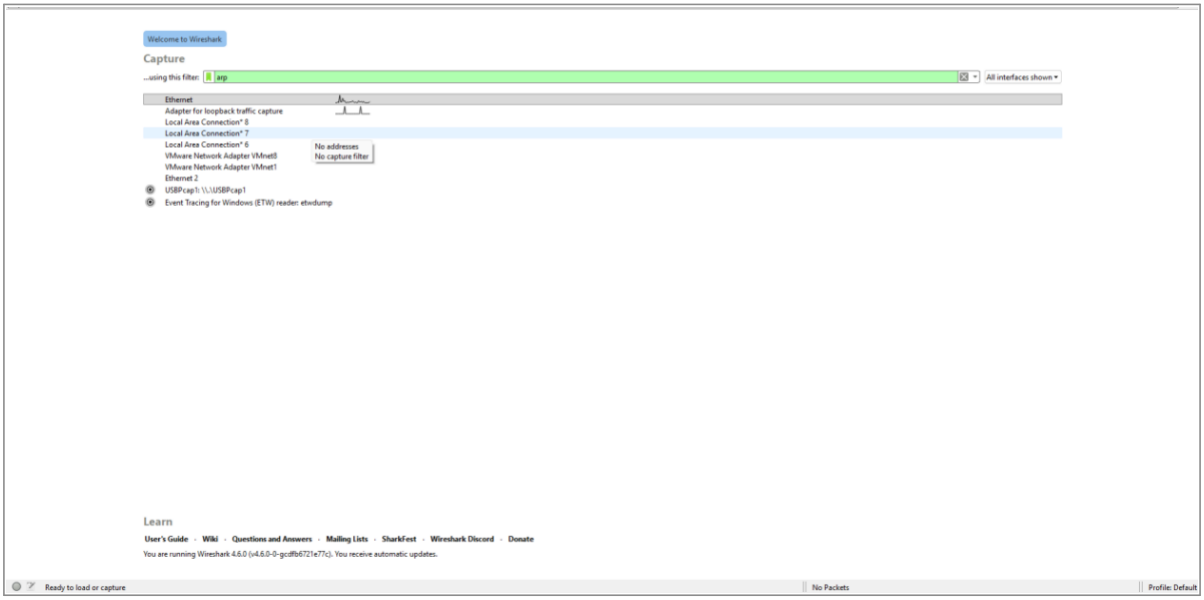
Wireshark welcome screen - list of interfaces (Ethernet selected)

No.	Time	Source	Destination	Protocol	Length	Info
455	4.029655	172.16.75.93	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
456	4.029786	172.16.75.93	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
457	4.034279	172.16.79.251	239.0.0.8	UDP	1078	39732 → 51721 Len=1036
458	4.039148	172.16.76.60	224.0.0.251	MDNS	119	Standard query response 0x0000 AAAA fe80::d989:6c0e:45e6:adac A 172.16.76.60
459	4.039433	172.16.79.249	239.0.0.8	UDP	1078	53696 → 51721 Len=1036
460	4.039433	fe80::d989:6c0e:45e…	ff02::fb	MDNS	139	Standard query response 0x0000 AAAA fe80::d989:6c0e:45e6:adac A 172.16.76.60
461	4.065317	172.16.79.248	239.0.0.8	UDP	1078	60790 → 51721 Len=1036
462	4.065976	CloudNetwork_79:1f:…	Broadcast	ARP	60	Who has 172.16.76.3? Tell 172.16.75.75
463	4.075041	172.16.79.251	239.0.0.8	UDP	1078	47989 → 51721 Len=1036
464	4.078653	fe80::cc8d:ce94:740…	ff02::fb	MDNS	139	Standard query response 0x0000 AAAA fe80::cc8d:ce94:740c:8676 A 172.16.76.144
465	4.078653	172.16.76.144	224.0.0.251	MDNS	119	Standard query response 0x0000 AAAA fe80::cc8d:ce94:740c:8676 A 172.16.76.144
466	4.078984	ea:55:ac:41:ed:03	Broadcast	ARP	60	Who has 172.16.76.6? Tell 172.16.76.144
467	4.079358	fe80::cc8d:ce94:740…	ff02::1:ff4b:dc37	ICMPv6	86	Neighbor Solicitation for fe80::305e:847a:194b:dc37 from ea:55:ac:41:ed:03
468	4.081102	172.16.79.249	239.0.0.8	UDP	1078	40958 → 51721 Len=1036
469	4.093706	fe80::490b:1406:77d…	ff02::fb	MDNS	139	Standard query response 0x0000 AAAA fe80::490b:1406:77db:69aa A 172.16.76.90
470	4.093706	172.16.76.90	224.0.0.251	MDNS	119	Standard query response 0x0000 AAAA fe80::490b:1406:77db:69aa A 172.16.76.90
471	4.093706	fe80::490b:1406:77d…	ff02::1:ff4b:dc37	ICMPv6	86	Neighbor Solicitation for fe80::305e:847a:194b:dc37 from 4c:82:a9:78:8c:e7
472	4.093706	CloudNetwork_78:8c:…	Broadcast	ARP	60	Who has 172.16.76.6? Tell 172.16.76.90
473	4.094340	CloudNetwork_79:20:…	Broadcast	LLC	60	S P, func=RNR, N(R)=64; DSAP NULL LSAP Individual, SSAP NULL LSAP Command
474	4.094565	fe80::75b7:683d:23d…	ff02::fb	MDNS	139	Standard query response 0x0000 AAAA fe80::75b7:683d:23d3:c11b A 172.16.76.112
475	4.094565	172.16.76.112	224.0.0.251	MDNS	119	Standard query response 0x0000 AAAA fe80::75b7:683d:23d3:c11b A 172.16.76.112
476	4.094701	172.16.76.112	224.0.0.251	MDNS	262	Standard query response 0x0000 A 172.16.76.112 AAAA fe80::2b:ec15:d25e:468a TXT
477	4.094701	172.16.76.112	224.0.0.251	MDNS	262	Standard query response 0x0000 A 172.16.76.112 AAAA fe80::2b:ec15:d25e:468a TXT
478	4.094810	172.16.76.112	224.0.0.251	MDNS	119	Standard query response 0x0000 AAAA fe80::75b7:683d:23d3:c11b A 172.16.76.112

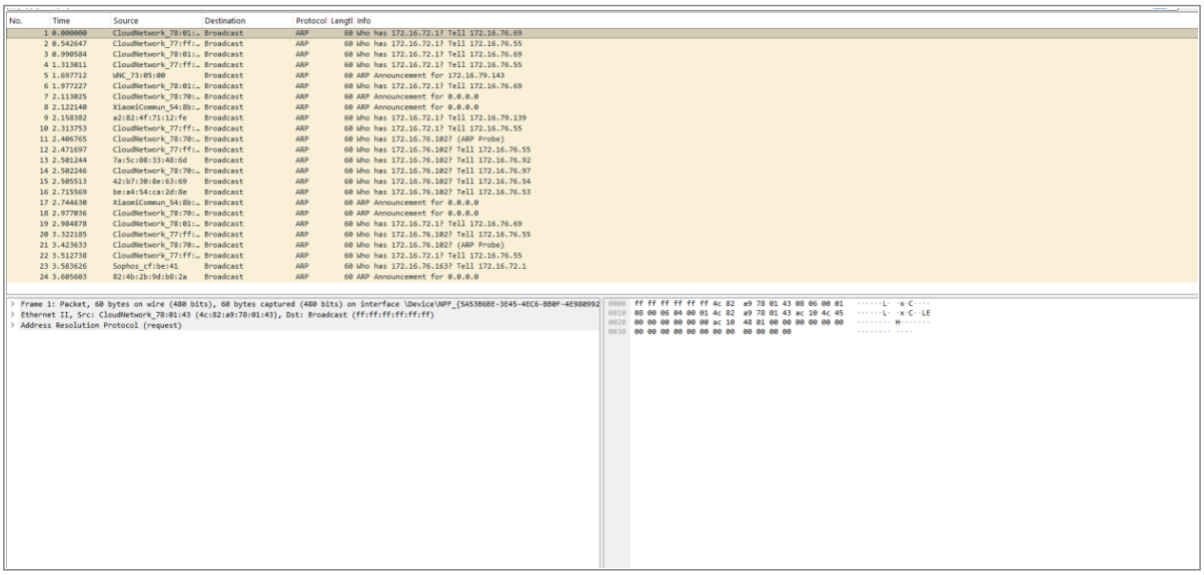
Packet List pane - captured packets (colours show the protocol)

```
01 00 5e 00 00 08 94 a6  7e a5 c2 1f 08 00 45 00   ··^····· ~·····E·
04 28 d4 25 40 00 01 11  b6 8b ac 10 4f fb ef 00   ·(·%@··· ····O···
00 08 ed 04 ca 09 04 14  0c 1f 0b 31 38 35 34 37   ········ ···18547
43 31 30 63 6f 6e 36 41  43 38 32 38 32 39 30 32   C10con6A C8282902
32 36 35 30 33 00 41 41  2d 41 38 2d 38 32 2d 34   26503·AA -A8-82-4
36 2d 31 38 2d 31 37 00  50 4f 43 4f 2d 58 36 2d   6-18-17· POCO-X6-
50 72 6f 2d 35 47 00 31  37 32 2e 31 36 2e 37 39   Pro-5G·1 72.16.79
2e 31 34 35 00 31 2c 33  2c 36 2c 31 35 2c 32 36   .145·1,3 ,6,15,26
2c 32 38 2c 35 31 2c 35  38 2c 35 39 2c 34 33 2c   ,28,51,5 8,59,43,
31 31 34 2c 31 30 38 00  00 00 00 00 00 00 00 00   114,108· ········
00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ········ ········
00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ········ ········
00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ········ ········
00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ········ ········
00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ········ ········
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00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ········ ········
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00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ········ ········
00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ········ ········
00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ········ ········
00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ········ ········
00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ········ ········
```

Packet Bytes pane - hexadecimal dump of a selected packet

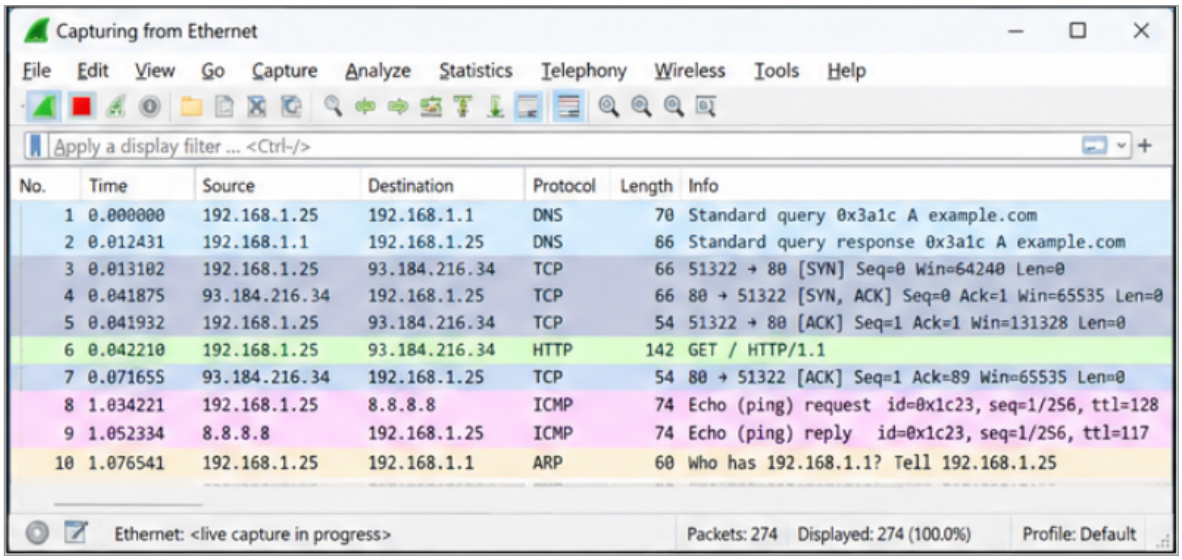


Capture filter *arp* entered before starting the capture

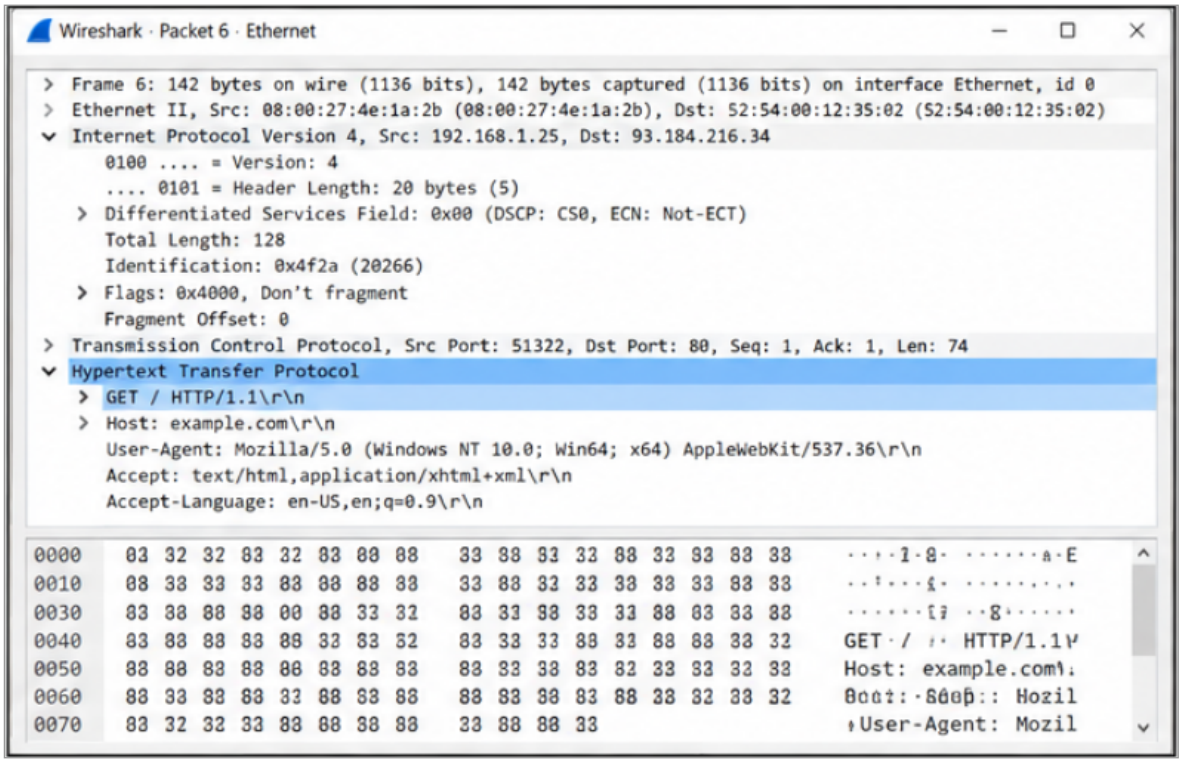


Captured ARP packets with the Frame, Ethernet II and ARP details

OUTPUT:

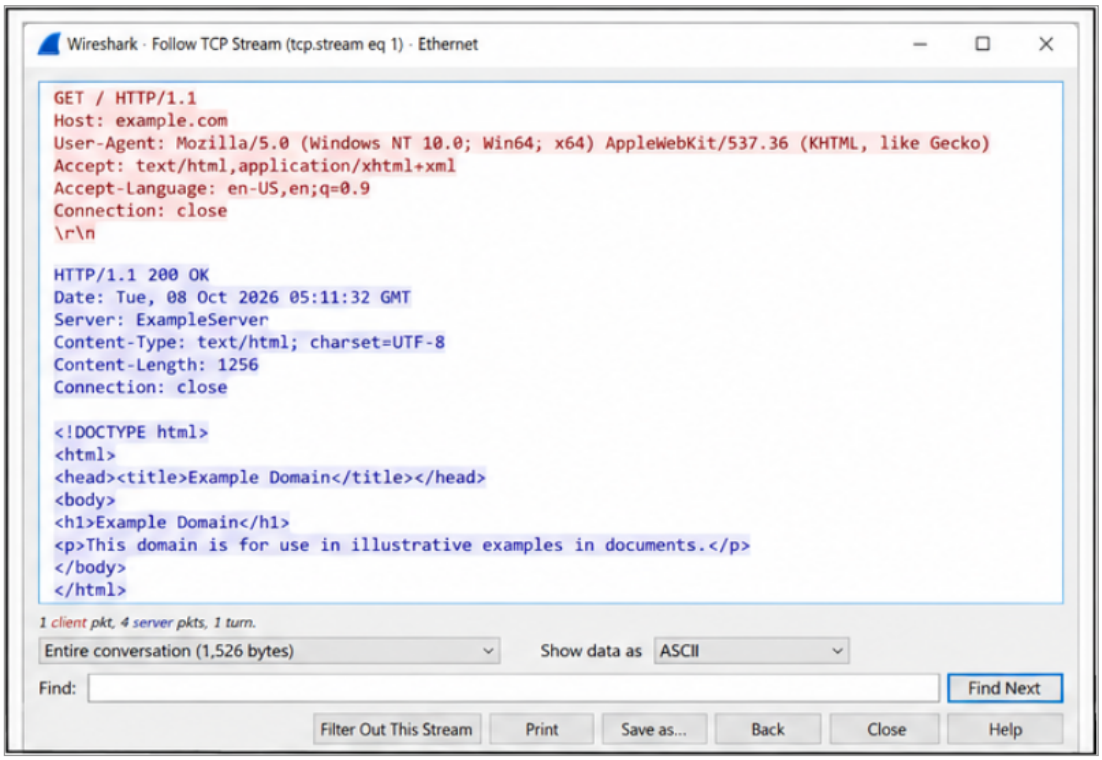


Capturing packets in Wireshark (packet list)

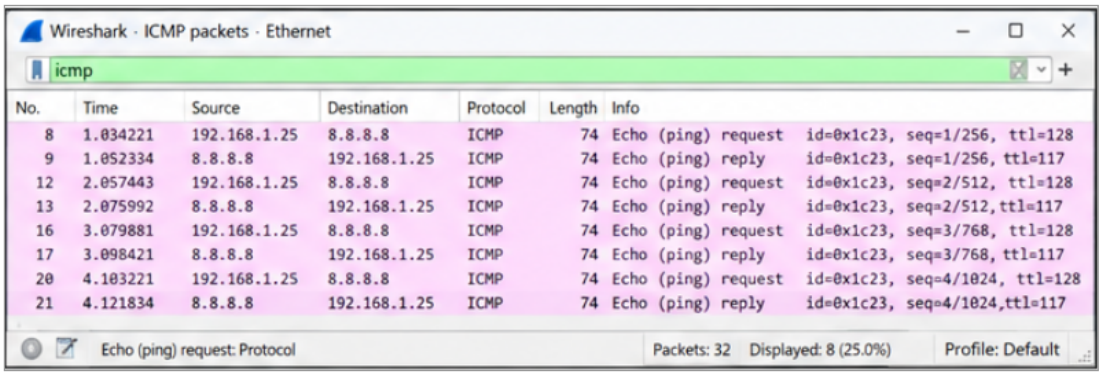


Packet details and packet bytes (HTTP GET request)

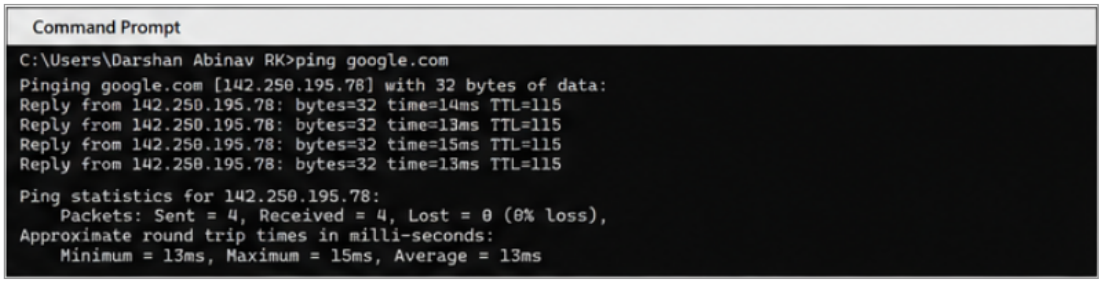
OUTPUT:



Follow TCP Stream (HTTP conversation)



Display filter (icmp) showing ICMP Echo Request / Reply packets



Command Prompt output (ping google.com)

RESULT:

Thus the features of Wireshark were studied. Packets were captured on the Ethernet interface using a capture filter, and the packet list, packet details and packet bytes of the captured SSDP, MDNS, ARP, ICMPv6, DNS, TCP, HTTP and ICMP packets were analysed.